

BSTT 527 Final

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Abstract

The Public Health system in the United States is changing as vaccine hesitancy increases and impacts the system that is already intact. By exploring cervical cancer data, the driver of this analysis was to explore and point out other features besides HPV in predicting cervical cancer and to share these findings with the public health community.

1 Introduction

Cervical cancer is the fourth most common cancer in women and almost all cases are linked to human papillomavirus (HPV). According to the World Cancer Research Fund and the WHO, among women, cervical cancer is ranked number 4 as of 2020 and accounts for about 7% of cancer in women. For all sexes, cervical cancer ranks at number 7 and accounts for 3.3%. Hispanic women have the highest rates of developing cervical cancer while black women have the highest death rates from cervical cancer (CDC). Women also living with HIV are more likely to get cervical cancer, and this impacts more women in middle to low-income countries.

Cervical cancer is developing tumors in the cells of the cervix which is the end of the uterus. Long-lasting HPV infections cause almost all cases of cervical cancer. According to the CDC, 4 of 5 women will be infected at least once in their life by age 50 and there are 13 types of HPV that can cause cervical cancer. HPV and cervical cancer can be significantly reduced through early diagnosis, prevention, vaccination, and treatments. In the United States, there are HPV vaccines given to women usually in their teens and are a set of three vaccines that help prevent certain strains of HPV. Even with the availability of this vaccine, vaccine hesitancy is growing in the United States and there are multiple studies that show an increase in HPV vaccine hesitancy from 2012 to 2020. With the Covid pandemic, there vaccine hesitancy for HPV became increasingly high, and according to the CDC, the vaccination rates of teens in the US for HPV dropped 12% for preteens and 28% for teenagers. According to the NIH from a study administered in Los Angeles, CA, about 20% had vaccine hesitancy for HPV, and more than 50% were concerned about the safety of the HPV vaccine. There are several studies that show within the vaccinated group

of HPV, cervical cancer is estimated at 87% less. Vaccine hesitancy is one of the largest public health challenges we face, the United States Government has even started a program each month promoting vaccines.

Another variable that increases the risk of cervical cancer is the long use of oral contraceptives. In 2002, the WHO linked long-term birth control to an increased risk of cervical cancer. There are studies that show the longer women use oral contraceptives, the more risk increases as time goes on. Still, specifically, oral contraceptives show an increased rate of developing cervical cancer at 10% if less than 5 years of oral contraceptive use, and upwards of 60% if they've been using contraceptives for 5-9 years (cancer.gov).

With the increasing hesitation in vaccines for HPV and their link to contraception use, I wanted to identify other possible features of cervical cancer and explore more machine-learning models associated with cervical cancer. The goal of this project is to explore tree-based Machine Learning Methods - Decision Tree and Gradient Boosting to predict the outcome of cervical cancer. Another goal is identifying the most important features associated with cervical cancer in addition to the factors known such as HPV and HIV. Another model I explored was Naive Bayes, a probabilistic classifier, on this dataset.

2 Data

The data file I used I found on Kaggle under ML Models for Prediction of Cervical Cancer. The dataset was collected at 'Hospital Universitario de Caracas' in Caracas, Venezuela. The dataset includes the following information demographics, sexual experience, and medical records of 858 patients. There are missing values associated with patients not answering all of the questions. This dataset focuses on the prediction of indicators/diagnoses of cervical cancer.

Most of the variables in this dataset are binary - representing if the patient has had STD - Sexual Transmitted Disease or Cancer. Then there are variables that are a sum or larger numbers including Age, Number of sexual partners, age at first sexual intercourse, the total number of years smoking, years that the patient has had hormonal contraceptives, the number of years the patient has had an IUD and the total number of sexual partners. To account for these variables, I used the 'scale' function in R to scale those variables that were total years, sums or age to normalize them.

To reduce the dataset even further, I looked at the Decision Tree that the overall data frame produced to see what features to prioritize. First I pulled the correlation matrix for the normalized variables to see if any were highly correlated. Other than age x how many years with hormonal contraception and age x age of first sexual intercourse there didn't seem to be much correlation between the normalized variables.

2.1 Tables

Variable Name	Variable Description	Variable Type
Age	Age of the woman	Numeric
Number of sexual partners	Total number of sexual partners	Numeric
Age - First sexual intercourse	Age of a woman	Numeric
Number of pregnancies	Total Pregnancies	Numeric
Smokes	Smokes(1) or not(0)	Binary
Smokes Years	Total Years	Numeric
Smokes Packs	Packs per year	Numeric
Hormonal Contraceptives	Whether on Contraceptives	Binary
Hormonal Contraceptives Years	Total Years	Numeric
IUD	Whether used or not	Binary
IUD Years	Years	Numeric
STD	Present or not	Binary
STD Total	Total Number	Numeric
STDs - condylomatosis	condylomatosis	Binary
STDs - cervical condylomatosis	cervical condylomatosis	Binary
STDs - vaginal condylomatosis	vaginal condylomatosis	Binary
STDs - vulvo-perineal condylomatosis	vulvo-perineal condylomatosis	Binary
STDs - syphilis	syphilis	Binary
STDs - PID	pelvic inflammatory disease	Binary
STDs - Herpes	genital herpes	Binary
STDs - molluscum contagiosum	molluscum contagiosum	Binary
STDs - AIDS	AIDS	Binary
STDs - HIV	HIV	Binary
STDs - Hep B	Hepatitis B	Binary
STDs - HPV	HPV	Binary
STDs	Number of diagnosis	Numeric
STDs	Time since first diagnosis	Numeric
STDs	Time since last diagnosis	Numeric
Diagnosis	Cancer	Binary
Diagnosis	Cervical intraepithelial neoplasia	Binary
Diagnosis	Presence of Human Papillomaviruses	Binary
Diagnosis	Presence any cancer, CIN, HPV	Binary
Hinselmann	Colposcopy	Binary
Schiller	Schiller Iodine test	Binary
Cytology	PaP smears test	Binary
Biopsy	cervical biopsy	Binary

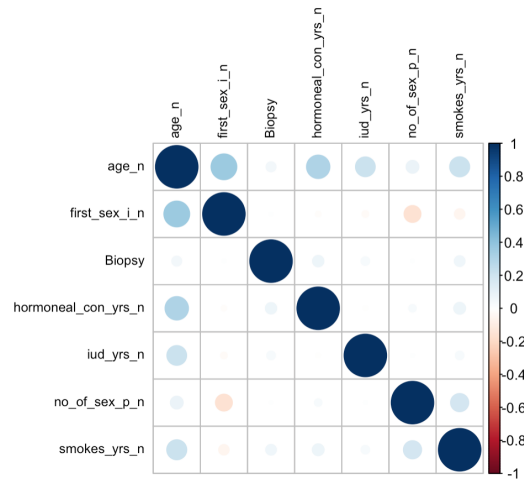


Figure 1: Correlation between Normalized Variables.

	vars	n	mean	sd	median	trimmed	mad	min	max	range
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
X	1	838	427.85	246.56	427.50	427.79	315.05	0.00	857.00	857.00
Age	2	838	26.81	8.53	25.00	26.02	7.41	13.00	84.00	71.00
No_of_sex_partner	3	838	2.51	1.59	2.00	2.32	1.48	1.00	28.00	27.00
First_sexual_intercourse	4	838	17.00	2.81	17.00	16.68	2.97	10.00	32.00	22.00
No_pregnancies	5	838	2.27	1.45	2.00	2.06	1.48	0.00	11.00	11.00
Smokes	6	838	0.15	0.35	0.00	0.06	0.00	0.00	1.00	1.00
Smokes_yrs	7	838	1.22	4.09	0.00	0.09	0.00	0.00	37.00	37.00
Smokes_packs_yr	8	838	0.45	2.23	0.00	0.02	0.00	0.00	37.00	37.00
Hormonal_Contraceptives	9	838	0.69	0.46	1.00	0.73	0.00	0.00	1.00	1.00
Hormonal_Contraceptives_years	10	838	2.36	3.81	0.58	1.52	0.86	0.00	30.00	30.00
IUD	11	838	0.10	0.30	0.00	0.00	0.00	0.00	1.00	1.00
IUD_years	12	838	0.45	1.82	0.00	0.00	0.00	0.00	19.00	19.00
STDs	13	838	0.09	0.29	0.00	0.00	0.00	0.00	1.00	1.00
STDs_number	14	838	0.15	0.52	0.00	0.00	0.00	0.00	4.00	4.00
STDs_condylomatosis	15	838	0.05	0.22	0.00	0.00	0.00	0.00	1.00	1.00
STDs_cervical_condylomatosis	16	838	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
STDs_vaginal_condylomatosis	17	838	0.00	0.07	0.00	0.00	0.00	0.00	1.00	1.00
STDs_vulvo_perineal_condylomatosis	18	838	0.05	0.22	0.00	0.00	0.00	0.00	1.00	1.00
STDs_syphilis	19	838	0.02	0.14	0.00	0.00	0.00	0.00	1.00	1.00
STDs_pelvic_inflammatory_disease	20	838	0.00	0.03	0.00	0.00	0.00	0.00	1.00	1.00
STDs_genital_herpes	21	838	0.00	0.03	0.00	0.00	0.00	0.00	1.00	1.00
STDs_molluscum_contagiosum	22	838	0.00	0.03	0.00	0.00	0.00	0.00	1.00	1.00
STDs_AIDS	23	838	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
STDs_HIV	24	838	0.02	0.14	0.00	0.00	0.00	0.00	1.00	1.00
STDs_Hepatitis_B	25	838	0.00	0.03	0.00	0.00	0.00	0.00	1.00	1.00
STDs_HPV	26	838	0.00	0.05	0.00	0.00	0.00	0.00	1.00	1.00
STDs_No_of_diagnosis	27	838	0.08	0.30	0.00	0.00	0.00	0.00	3.00	3.00
Dx_Cancer	28	838	0.02	0.14	0.00	0.00	0.00	0.00	1.00	1.00
Dx_CIN	29	838	0.01	0.10	0.00	0.00	0.00	0.00	1.00	1.00
Dx_HPV	30	838	0.02	0.14	0.00	0.00	0.00	0.00	1.00	1.00
Dx	31	838	0.03	0.16	0.00	0.00	0.00	0.00	1.00	1.00
Hinselmann	32	838	0.04	0.20	0.00	0.00	0.00	0.00	1.00	1.00
Schiller	33	838	0.09	0.28	0.00	0.00	0.00	0.00	1.00	1.00
Citology	34	838	0.05	0.22	0.00	0.00	0.00	0.00	1.00	1.00
Biopsy	35	838	0.06	0.25	0.00	0.00	0.00	0.00	1.00	1.00
age_n	36	838	0.00	1.00	-0.21	-0.09	0.87	-1.62	6.70	8.32
no_of_sex_p_n	37	838	0.00	1.00	-0.32	-0.12	0.93	-0.95	16.05	17.00
first_sex_i_n	38	838	0.00	1.00	0.00	-0.11	1.05	-2.49	5.33	7.82
smokes_yrs_n	39	838	0.00	1.00	-0.30	-0.28	0.00	-0.30	8.75	9.04
hormoneal_con_yrs_n	40	838	0.00	1.00	-0.47	-0.22	0.23	-0.62	7.26	7.88
iud_yrs_n	41	838	0.00	1.00	-0.24	-0.24	0.00	-0.24	10.17	10.42

41 rows

Figure 2: Descriptive Statistics on Dataset.

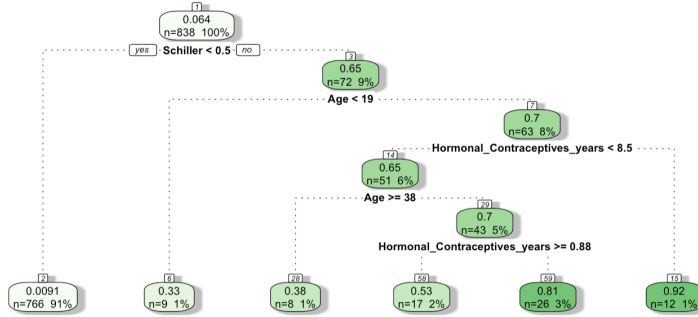


Figure 3: Decision Tree for Feature Selection.

3 Data Exploration

The target variable for this dataset is Biopsy which is a Binary. From the figure above we can see that the Median for most of these variables is 0.0, since most of the variables are Binary. To focus on the important features and reducing the dataset, I used an R package tree, rattle, rpart to plot a Decision Tree to pick out the most meaningful features.

From the Decision Tree below, the features that are most important are Schiller, Age and Years of Hormonal Contraceptives. The variable Schiller represents a Schiller Iodine test which is a medical test in which iodine solution is applied to the cervix in order to diagnose early cervical cancer. Then there is a breakout of Age, it seems $Age \leq 19$ or $Age \geq 38$ If the patient has used hormonal contraceptives for years. The number of years of hormonal contraceptives seems to look like around 8.5 years. This decision tree is quite unique, since I was expecting HPV and/or HIV in this decision tree.

The paths to a positive Biopsy include:

1. A positive Schiller test
2. A negative Schiller test + younger than 19 years old
3. A negative Schiller test + hormonal contraceptives less than 8.5 years + younger than 38 years of age
4. A negative Schiller test + hormonal contraceptives less than 8.5 years + older than 38 years of age + less than 0.88 years on hormonal contraceptives
5. A negative Schiller test + hormonal contraceptives less than 8.5 years + older than 38 years of age + more than 0.88 years on hormonal contraceptives
6. A negative Schiller test + older than 19 years old + hormonal contraceptives more than 8.5 years

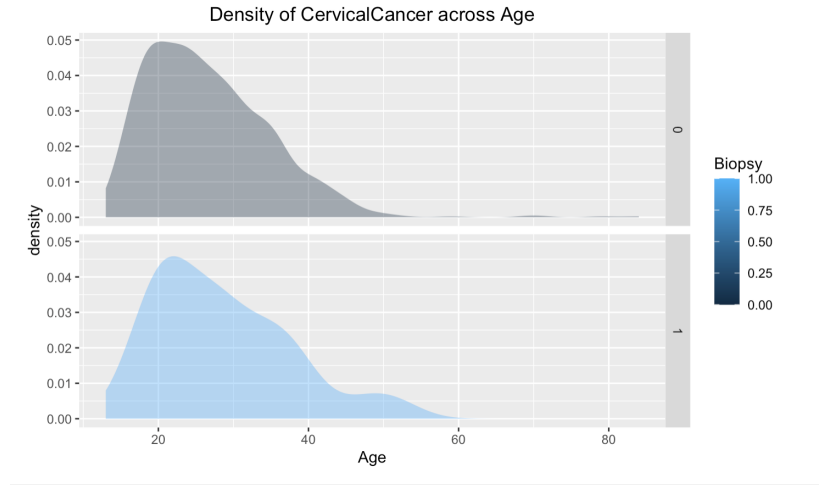


Figure 4: Density of Cervical Cancer Across Age.

The first figure above, figure 4 represents the density of cervical cancer across age. The top graph shows Biopsy 0, so cancer isn't present while the bottom Biopsy 1 so cancer is present. Though the densities have similar shape, the bottom plot shows a second hump which is almost bimodal where the age is around 50 years old. The first peak is later than the peak without cervical cancer, by a couple of years but still early 20's.

The second figure below, figure 5 represents the density of cervical cancer across the years on hormonal contraceptives. Though the densities have similar shape, the bottom plot shows not nearly as high of a peak in the beginning and looks like there is a larger tail being stretched to the right. On the bottom graph there are even three peaks you could say, with the right two peaking at 8-9 years of contraceptive use and around 15 years of contraceptive use.

The correlation plot below, figure 6 represents the correlation among variables in the reduced dataset. The correlations that are apparent are Schiller x Biopsy, Citology x Biopsy, Dx HPV x Biopsy, Number of Years on Hormonal Contraceptives x Biopsy, and Dx x Biopsy. The variable Dx - It indicates the presence any one among cancer, CIN and HPV.

4 Data Preparation - Testing and Training Data

For the analysis - there was a 90/10 split on the data for training and testing. Since there were only 54 / 838 positive Biopsy results, I went with a 90/10 over an 80/20. Initially I started with an 80/20 split but realized the dataset wasn't evenly distributed and to keep more in the training set.

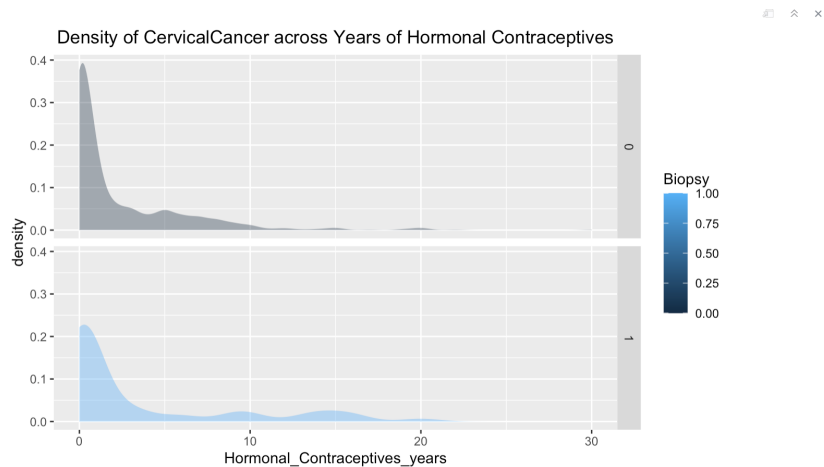


Figure 5: Density of Cervical Cancer Across Years of Hormonal Contraception.

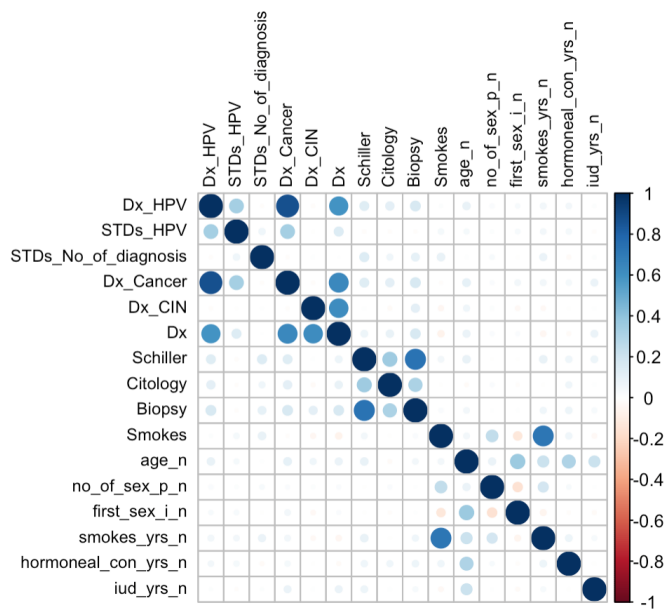


Figure 6: Correlation of Reduced Matrix.

5 Methodology

5.1 Decision Tree

Decision trees are a supervised (labeled data) machine learning algorithm that can be used for both classification and regression. The advantages as seen above include high interpretability and can be considered close to if-else statements. The disadvantages are overfitting and a small change in the dataset can lead to a different decision tree. A decision tree is structured with the outcome variable at the top with nodes below that breakout into more nodes. For the decision tree above, there weren't any restrictions like on max depth or splitting the data.

The output of the decision tree guided the analysis on which features and variables to focus on in analyzing cervical cancer data. From the Decision tree, I decided to focus on Schiller, Age and Number of Years of Hormonal Contraceptives which later fed into Logistic Regression, Naive Bayes and Gradient Boosting Model.

5.2 Logistic Regression

Logistic Regression. The first model I ran was Logistic Regression which is a regression model where the outcome variable has categorical values such as 0 and 1. Logistic Regression is a type of Classification algorithm with the response variable being categorical - in this case Biopsy at 0 or 1. The point of logistic regression is to find a relationship between features and probability of a particular outcome. In the figures I ran two logistic regression models, first an overall logistic regression model using all the variables and second a logistic regression model using only 3 features: Schiller, age and years of hormonal contraceptives.

From the AIC values of the models above, we see an AIC of 184 for the broader model and for the one with 3 features an AIC of 179.43. Since there isn't a notable differences between a logistic regression on all the variables vs the 3 variables from the Decision tree it is time to start considering a different model.

From the summaries of logistic regression, its apparent that Schiller is one of the strongest predictive variables and has the lowest p-value. We can interpret this as any one unit change in Schiller will increase the odds of getting a positive Biopsy by 5.518 and the p-value shows it is significant in determining a positive Biopsy result. For each unit increase in years of hormonal contraceptives will increase the odds of getting a positive Biopsy by 0.15.

Interpreting the Residuals vs the Predicted Values - the residuals hover around the zero line and fall both above and below. There are a few residuals that stand out which means there are a few outliers.

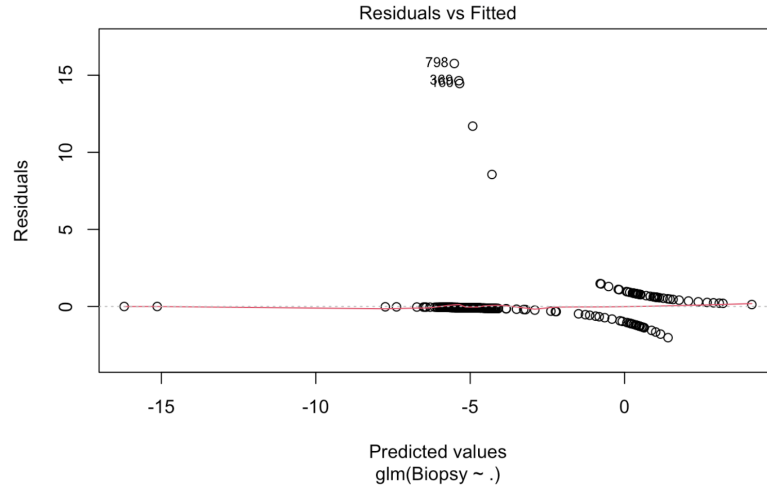


Figure 7: Logistic Regression - AIC:184.4449.

5.3 Naive Bayes

Naive Bayes algorithm is a classification algorithm based on Bayes' Theorem of Probability. The main assumption is that the features are independent. The main driver behind using Naive Bayes over other classification models, is to capture and predict rare events. Out of the target variable Biopsy, there are 54 positive out of 858 observations so this is considered a rare event. Naive Bayes is a computationally fast model to run on test data, which here is split 90/10. When the independence of variables hold, Naive Bayes performs better than logistic regression and other classification models. Another strong trait of Naive Bayes is the performance with categorical variables compared to numeric and since the dataset is mostly comprised of binary variables, this would be the best model to consider.

The accuracy of the overall model is pretty surprising on a training dataset with over 90 % accuracy according to the Confusion Matrix. The second model including the features from the Decision Tree, is almost at 95 % accuracy which is extremely high for a classification model with both binary and numeric variables.

In Figure 11 and 12 showing the two Naive Bayes models for this analysis - this shows the Confusion Matrix per each model. The sensitivity for both models represents how well the model is at detecting positive Biopsy result which is rare in our dataset. The first model is almost at 91% and the second model is almost 96% which is extremely high. The specificity measures how exact the assignment to positive Biopsy result which is almost 92% and then 85% for the second model.

```

Call:
glm(formula = Biopsy ~ ., family = binomial, data = X_r)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.8032  -0.1179  -0.1073  -0.0966   3.3223

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    -5.17126    0.48523  -10.657  <2e-16 ***
Dx_HPV           1.21662    4.83881    0.251   0.8015
STDs_HPV       -12.99125   1012.43055  -0.013   0.9898
STDs_No_of_diagnosis  0.03708    0.50723    0.073   0.9417
Dx_Cancer       1.44588    5.29791    0.273   0.7849
Dx_CIN          5.11360    2.98340    1.714   0.0865 .
Dx_HPV.1         NA         NA         NA      NA
Dx              -0.82822    2.82010   -0.294   0.7690
Schiller        5.51822    0.55873    9.876  <2e-16 ***
Citology        0.69524    0.58144    1.196   0.2318
Smokes         -0.09646    0.89214   -0.108   0.9139
age_n          -0.36899    0.25550   -1.444   0.1487
no_of_sex_p_n   0.14053    0.22542    0.623   0.5330
first_sex_i_n   0.29758    0.28940    1.028   0.3038
smokes_yrs_n    0.15367    0.25391    0.605   0.5450
hormoneal_con_yrs_n 0.18575    0.19673    0.944   0.3451
iud_yrs_n       0.03760    0.16462    0.228   0.8193
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

Figure 8: Summary of Logistic Regression.

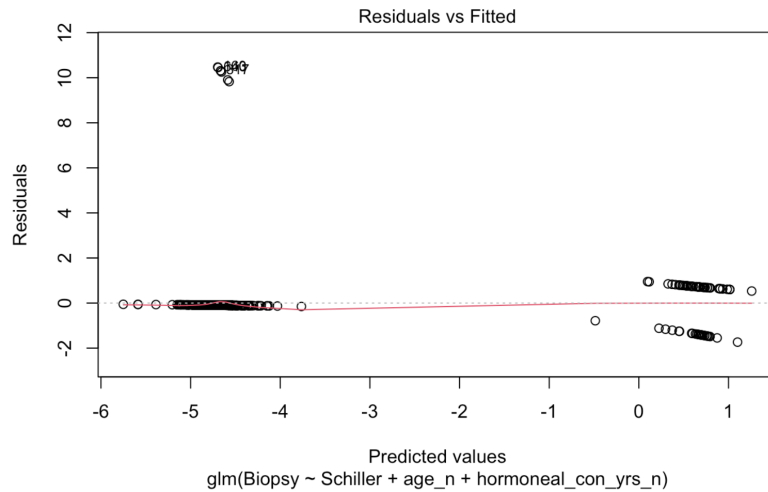


Figure 9: Logistic Regression - AIC: 179.4304.

Call:

```
glm(formula = Biopsy ~ Schiller + age_n + hormoneal_con_yrs_n,
     family = binomial, data = X_r)
```

Deviance Residuals:

Min	1Q	Median	3Q	Max
-1.6662	-0.1416	-0.1354	-0.1257	3.0682

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.7026	0.3809	-12.347	<2e-16 ***
Schiller	5.3561	0.4611	11.616	<2e-16 ***
age_n	-0.1554	0.1821	-0.853	0.394
hormoneal_con_yrs_n	0.1575	0.1739	0.906	0.365

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 400.58 on 837 degrees of freedom
 Residual deviance: 171.43 on 834 degrees of freedom
 AIC: 179.43

Number of Fisher Scoring iterations: 7

Figure 10: Summary of Logistic Regression.

```
model <- naiveBayes(Biopsy ~ ., data = train_f2)
pred <- predict(model, train_f2[, -10])
confusionMatrix(pred, as.factor(train_f2$Biopsy))
```

Confusion Matrix and Statistics

```

      Reference
Prediction 0  1
0    641  4
1     66 43

Accuracy : 0.9072
95% CI : (0.8842, 0.9269)
No Information Rate : 0.9377
P-Value [Acc > NIR] : 0.9996

Kappa : 0.5085

McNemar's Test P-Value : 3.079e-13

Sensitivity : 0.9066
Specificity : 0.9149
Pos Pred Value : 0.9938
Neg Pred Value : 0.3945
Prevalence : 0.9377
Detection Rate : 0.8501
Detection Prevalence : 0.8554
Balanced Accuracy : 0.9108

```

Figure 11: Base Naive Bayes model.

```
model1 <- naiveBayes(Biopsy ~ Schiller + age_n + hormoneal_con_yrs_n, data = train_f2)
pred1 <- predict(model1, train_f2[, -10])
confusionMatrix(pred1, as.factor(train_f2$Biopsy))
```

Confusion Matrix and Statistics

```

      Reference
Prediction 0  1
0    676  7
1     31 40

Accuracy : 0.9496
95% CI : (0.9315, 0.9641)
No Information Rate : 0.9377
P-Value [Acc > NIR] : 0.0973788

Kappa : 0.6519

McNemar's Test P-Value : 0.0001907

Sensitivity : 0.9562
Specificity : 0.8511
Pos Pred Value : 0.9898
Neg Pred Value : 0.5634
Prevalence : 0.9377
Detection Rate : 0.8966
Detection Prevalence : 0.9058
Balanced Accuracy : 0.9036

```

Figure 12: Naive Bayes model with 3 features.

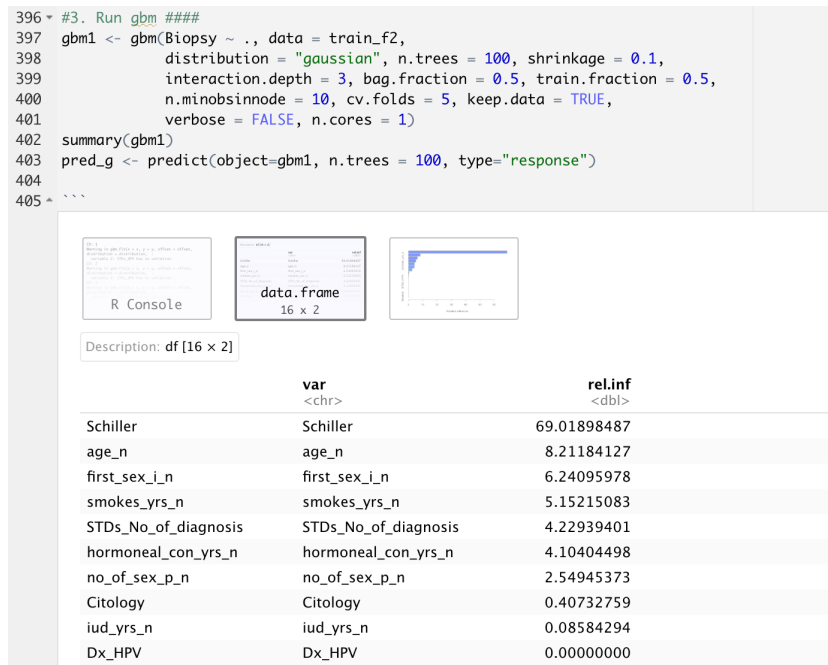


Figure 13: Relative Influence GBM Model 1.

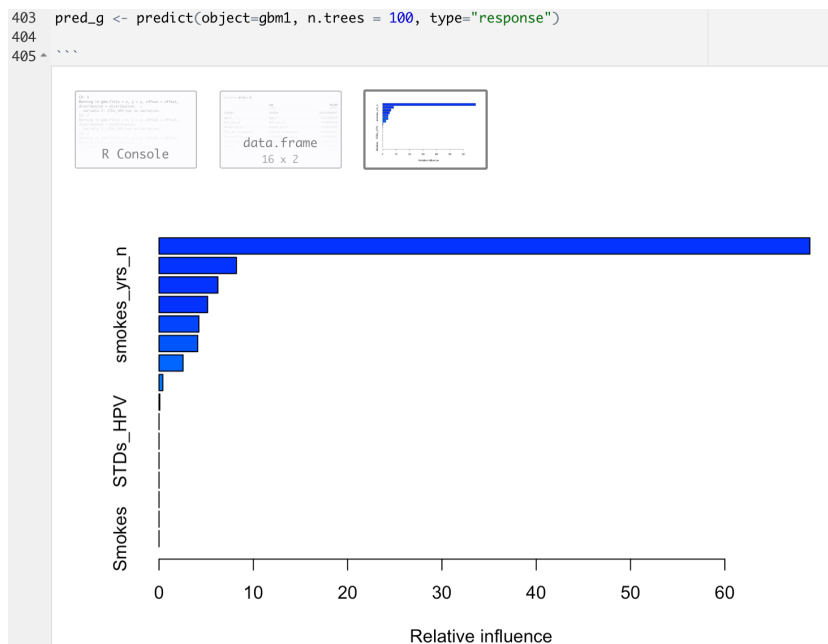


Figure 14: Relative Influence GBM Model 1.

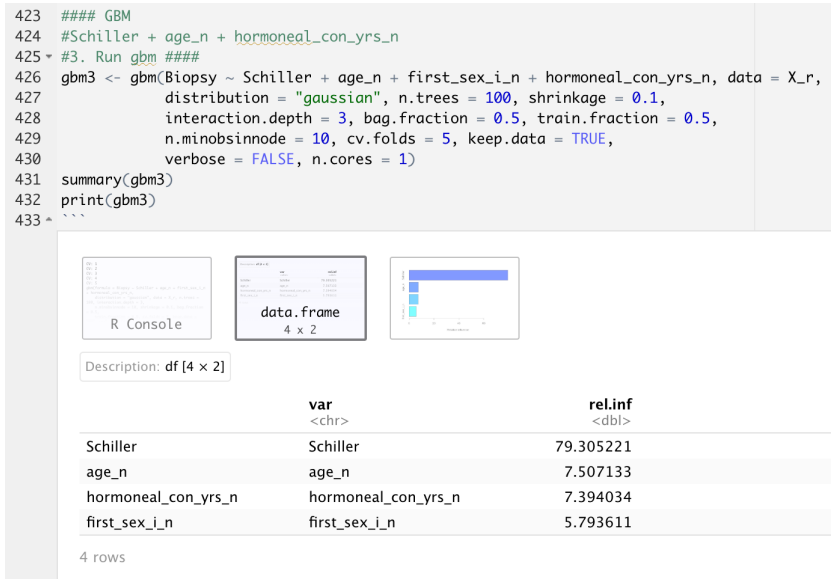


Figure 15: Relative Influence GBM Model 2.

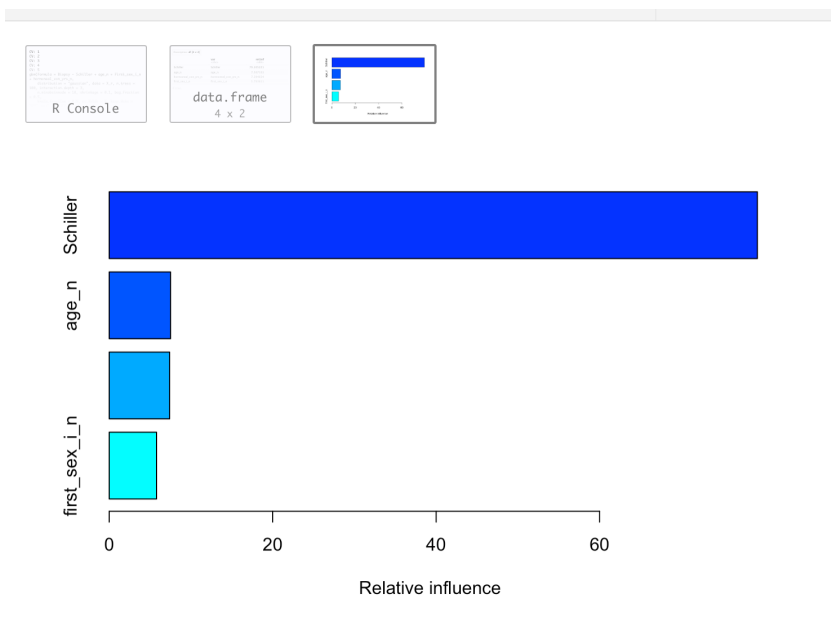


Figure 16: Relative Influence GBM Model 2.

```

CLASSIFICATION: If so, use a 2 level factor as your outcome column.
Stochastic Gradient Boosting

754 samples
16 predictor

No pre-processing
Resampling: Cross-Validated (5 fold, repeated 5 times)
Summary of sample sizes: 603, 603, 603, 604, 603, 603, ...
Resampling results across tuning parameters:

n.trees  RMSE      Rsquared  MAE
200      0.1824460  0.4520010 0.07496303
250      0.1844848  0.4427711 0.07687452
300      0.1853405  0.4393972 0.07779194
350      0.1864227  0.4337790 0.07915606
400      0.1872290  0.4298479 0.08012415
450      0.1879585  0.4271378 0.08094372
500      0.1886538  0.4233616 0.08150648

Tuning parameter 'interaction.depth' was held constant at a value of 5
Tuning parameter 'shrinkage'
was held constant at a value of 0.05
Tuning parameter 'n.minobsinnode' was held constant at a value of 10
RMSE was used to select the optimal model using the smallest value.
The final values used for the model were n.trees = 200, interaction.depth = 5, shrinkage = 0.05
and n.minobsinnode = 10.
[1] 0.1561447

```

Figure 17: Correlation of Reduced Matrix.

5.4 GBM: Generalized Boosting Model

According to R documentation GBM is an implementation of extensions to Freund and Schapire's AdaBoost algorithm and Friedman's gradient boosting machine. Includes regression methods for least squares, absolute loss, t-distribution loss, quantile regression, logistic, multinomial logistic, Poisson, Cox proportional hazards partial likelihood, AdaBoost exponential loss, Huberized hinge loss, and Learning to Rank measures (LambdaMart). Originally developed by Greg Ridgeway.

GBM: Gradient Boosting Model - is a supervised Machine Learning algorithm used for both classification and regression problems. The technique uses weak learners to produce a strong model for classification and regression. It takes all of the outputs of the weak learners to create a model with higher accuracy.

AdaBoost: The logic behind AdaBoost is that it fits models on different weighted training data. First it gives equal weight to each observation and then cycles through the learners and gives higher weight to the observations predicted incorrectly. Since it is iterative it goes until a number of models is reached or a threshold of accuracy. AdaBoost starts by training a decision tree where it starts with equal weighted observations and then continues to more decision trees.

One of the most prominent properties of GBM is the ability to model variable importance. The output of the model is a table ranking the variables by importance followed by a bar graph displaying the variables with the highest relative influence. The table and plots in Figure 13 - 16 display the most influential

features for the predictive model.

From the RMSE above for the GBM model the RMSE is used as a measure that tells us the average distance between the predicted and actual values in the dataset. The lower the RMSE the better the model is fitting the dataset. With the output of figure 17 showing that as the number of trees increases the RMSE also increases, shows that we should either use a lower model of trees or a different model for this dataset.

6 Conclusion

From the findings above, it is very apparent that the Naive Bayes model performed the best on the dataset. The feature Schiller test seems like the strongest driver for the predictive model on cervical cancer. Even though the Schiller test is much stronger than the other features when it comes to influence it does show that Age, Number of Years on Hormonal Contraceptives, Number of years the patient has smoked, How old the person was during their first sexual intercourse and even the number of sex partners do have an impact as well. After running another model using Naive Bayes on the features from the output of GBM, the accuracy was within 0.01 of the model in Figure 12, so there wasn't much of a difference with adding 4 additional features.

The use of Decision Trees in the beginning to breakdown the dataset with all of the binary variables to show which variables to focus on was the best way to start exploring. However, I should've implemented GBM sooner in the Data Exploring rather than after running Logistic and Naive Bayes models. While comparing Logistic Regression vs Naive Bayes, the Logistic Regression isn't as likely to overfit the data as it is a linear model, but it doesn't handle missing data well. The Naive Bayes model was better at fitting this cervical cancer dataset because the backbone of Naive Bayes is predicting probability and classification. The only downside of Naive Bayes is ensuring all of the variables are independent, but I picked variables in the Naive Bayes model purposely excluding HPV to avoid this issue.

6.1 Discussion

The vast majority of research on cervical cancer is done with early screenings, vaccine prevention of HPV and general reproductive health. But as of 2022, the rates of deadly cervical cancer are continuing to grow and are more deadly for both hispanic and black women. Since vaccine hesitancy for HPV is over 50%, it would be extremely impactful to have research on women where there are vaccinated people in the dataset as well or research done to help reduce vaccine hesitancy.

6.2 Limitations

The cervical cancer dataset hardly had any positive Biopsy results for cervical cancer. The total was 54/838 which isn't a high number to predict and is considered a rare event. Because the lack of positive Biopsy - there were limited models to use on binary data but Naive Bayes did perform extremely well.

Another limitation was understanding the dataset completely. Although the outcome variable is Biopsy I explored other research and papers suggesting that a positive result in any of the last 4 columns suggested the patient had cervical cancer but that wasn't noted on the UCI ML database, so I decided to stick with the positive Biopsy result as stated on UCI's website. Another route I could have explored was removing HPV diagnosis from the dataset early on to see if other features were more prominent - even though Schiller not HPV was the most prominent variable.

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